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Engineering Leadership
AI & Data Platform Team
RedChip Companies, Inc.
Orlando, FL, USA (Remote)

Dear Engineering Leadership and AI Platform Team,

RedChip's strategic expansion of the RedChat platform into a real-time financial intelligence assistant transforms how investors extract insights from complex SEC filings, market data, and small-cap disclosures. Powering this conversational research loop requires resilient, queue-backed data pipelines, high-precision hybrid retrieval across numerical financial statements, and low-latency API infrastructure. Having engineered production-grade data ingestion pipelines and custom retrieval engines in **Python**, **FastAPI**, and **PostgreSQL**, my background directly addresses the challenges of turning massive unstructured regulatory filings into sub-second assistant workflows.

During my software engineering tenure at **LexoLegal**, I architected asynchronous REST APIs in **FastAPI** and **PostgreSQL** behind an **Nginx** proxy, building autonomous assistants with **sub-2s latency** via **Pydantic AI** tool-calling loops. To maintain real-time document synchronization across 25 High Courts and 743 District Courts, I automated daily ingestion pipelines via **Celery** workers and **Selectolax** DOM extractors, executing batch **PostgreSQL** inserts with atomic **Redis** distributed locks to eliminate race conditions. Furthermore, in **ncert-rag-evals** and **vectorless-rag**, I implemented custom hybrid search in **SQLite FTS5** and **ChromaDB** via Reciprocal Rank Fusion (RRF), proving an empirical **86.9% Recall@5 in 8.9ms on CPU** (a **99.8% latency reduction**), while streaming document windows in **PyMuPDF** under a strict **under-200MB memory ceiling**.

Financial research systems demand absolute numerical precision and high reliability over black-box abstractions. Across my projects, every system is reinforced with empirical evaluation and comprehensive automated test suites, including 92 **Pytest** suites in **ncert-rag-evals** with paired McNemar statistical significance testing (**p = 0.0021**) and 151 tests in **doubtless** with distributed tracing across **OpenTelemetry** and **Logfire**. This disciplined engineering ensures data pipelines process filings without worker starvation or silent data loss.

I am eager to contribute immediately to RedChip's AI and data platform as a Software Engineer. Based in India, I am fully equipped for remote contract work, comfortable collaborating across international timezones, and welcome the opportunity to discuss the technical direction of RedChat.

Sincerely,

Aman Verma